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A novel coarse-to-fine computational method for three-dimensional landmark detection to perform hard-tissue cephalometric analysis

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Abstract

Cephalometric analysis has an important and essential role to treat the patients with craniofacial and dentofacial deformities. Cephalometric analysis is a relationship of human geometry which can be quantified and derived from the linear and angular measurements. To treat any patient, such analysis is required to be performed on the Head X-ray image of the patient. The objective of the proposed work is to detect cephalometric landmarks automatically on CT (computational tomography) images. Twenty cephalometric landmarks were automatically localized on 100 CT scans using hybrid coarse-to-fine computational method. The mean error for landmark detection was computed as 2.88 mm and standard deviation of 1.85 mm. The highest detection rate for cephalometric landmarks was received as 100% for Nasion landmark under 4-mm error and the highest detection rate was received as 99% for Nasion landmark under 3-mm error. The less number of datasets were used for the training and higher number of datasets were used for the testing. Compared to the literature methods, our method used higher number of datasets to demonstrate the accuracy of the proposed method.

KEYWORDS
3D dental analysis, cephalometric analysis, CT, landmark, landmark detection

1 | INTRODUCTION

Human may have the craniofacial and oral deformities which need to be corrected by orthodontists by performing required Orthodontic and Dentofacial Orthopaedic treatment (Kharbanda, 2019; Midtgård et al., 1974). Cephalometric analysis has an important and essential role to treat such patients. Cephalometric analysis is a relationship of human geometry which can be quantified and derived from the linear and angular measurements (Chengoden et al., 2023; Singh et al., 2023). To treat any patient, such analysis is required to be performed on the Head X-ray image of the patient (Baldini et al., 2022).

Cephalometric analysis was started in around 1948 (Downs, 1948). Initially, the analysis was performed on the X-ray tracings without using computers which was very difficult to measure the precise distances and angles (Friedland, 1998; Nessi et al., 1993). However, later on, with the use of computer, the same analysis become easy to perform and record the measurements (Hashash et al., 2021; Yoon et al., 2001). But this was boring and tiring due to the manual efforts. It was taking much time to perform the analysis. Therefore, with more computational power, computationally automatic methods were evolved for performing cephalometric analysis automatically (Grau et al., 2001; Leonardi et al., 2008; Song et al., 2020; Yue et al., 2006). However, while with the more progression in technology, CT and CBCT images were also acquired (Berco et al., 2009; Olszewski et al., 2006). It makes the visualization of the 3D craniofacial structure easy to present and analyse (Kochhar et al., 2021). However, again the manual landmark plotting on 3D craniofacial images using 2D monitor become difficult (Kim et al., 2022). Therefore, now there is a requirement of automation in the landmark detection method (Saab & Jaafar, 2021; Sayour et al., 2022; Shen et al., 2021).

Cephalometric landmark detection using three-dimensional images are beneficial in terms of non-overlapping of craniofacial structures, visibility of clear spatial coordinates and visibility of anatomical geometry for asymmetric patients (Chengoden et al., 2023; Song et al., 2019). However, three-dimensional images exposes the patients with higher radiation doses which is not recommendable until cost benefit ratio is justified (Hammoud et al., 2022; Saab et al., 2021).

There are a lot of methods available in the literature for the automatic detection of the cephalometric landmarks. However, no method is established yet and still there is a need of new computational methods for the automatic landmark localization method. In the literature, several types of methods are there. Starting from model based (Shahidi et al., 2014) and knowledge-based methods (Arisdakessian et al., 2020; Gupta et al., 2022), currently mostly methods are based on machine learning (ML) based (Kang et al., 2021; Yun et al., 2020) and deep learning (DL) based (Lee et al., 2019). At a very recent methods are based on combined approaches, as this article is also adopted the similar ensemble approach. In addition, researchers are working to reduce the amount of radiation exposure to the patient for acquiring CT images (Kissel et al., 2021).

In the literature, there are many types of methods started from 2014 by model-based approach (Shahidi et al., 2014). It had detected 14 cephalometric landmarks on 20 CBCT images with 63.57% detection rate under 3-mm error range. A knowledge-based method was proposed for the cephalometric landmark detection (Gupta et al., 2022) followed by cephalometric analysis (Arisdakessian et al., 2020) over 30 CBCT images where mean error for landmark detection was reported as 2.01 mm and 82.67% detection rate was achieved for the landmarks under 3 mm error range. During the automatic cephalometric analysis, maximum average error in the measurements was found as 2.63 mm. Later on, a knowledge-based approach was published in 2015 (Gupta et al., 2022) to demonstrate automatic landmark detection on 30 CBCT images. This experiment was performed on 30 samples with 20 hard tissue landmarks. The same group had also demonstrated the automatic cephalometric analysis using the said approach (Arisdakessian et al., 2020; Gerges et al., 2021; Helwan et al., 2021) and results were found satisfactory. Montúfar et al. (2018) has demonstrated the experiment for the automatic localization of 18 cephalometric landmarks over 24 sample images and they were received 3.64 mm average error. Lee et al. (2019) has demonstrated the experiment for the automatic localization of seven cephalometric landmarks over seven sample images and they were received 1.5 mm average error. The received average error was very less compared to the other studies in the literature. Although the dataset used in this study was very less compared to the other studies in the literature. Yun et al. (2020) has demonstrated the experiment for the automatic localization of 93 cephalometric landmarks over 26 sample images and they were received 3.63 mm average error.

These method were improved a lot by another authors in further years. Recently, Kang et al. (2021) has demonstrated a reinforcement learning method for 3D landmark detection where 16 landmarks were plotted on eight testing datasets to demonstrate the accuracy of the method. The mean average error was found significantly low as 1.96 mm. However, the testing dataset was significantly low in this experiment. Dot et al. (2022) has performed DL-based experiment for automatic cephalometric landmark detection. Thirty-three landmarks were plotted automatically on 38 testing images to demonstrate the accuracy of the method. The methods shown in the literature are very progressive. However, still there is no defined method which can give the promising accuracy or clinical use for the clinical practice. Therefore, more research work is required in this direction (Abbas et al., 2021; Tarhini et al., 2022). Therefore, we have proposed a hybrid coarse-to-fine method to annotate 3D cephalometric landmarks on CT images with higher sample data.

The rest of this article is structured as follows: Section 2 describes the materials and methods used in this work. In Section 3 results are given followed by the discussions in Section 4. Finally, conclusion is given in Section 5.

2 | MATERIALS AND METHODS

ever, the proposed method had deal with this to overcome the problem.

A total of 20 cephalometric landmarks were chosen for the automatic localization of cephalometric landmarks. These landmarks were hard tissue landmark. No soft-tissue landmark was chosen for the automatic localization. The selected landmarks are very common in nature for the automatic detection in cephalometric analysis. Table 1 shows the 20 cephalometric landmarks which were identified for our experiment. The definitions of the landmarks are given in Table 1.

2.1 | Automatic 3D landmark detection

The volume of each patient was very large due to the inclusion of neck portion of body. Therefore, we have removed the neck portion from the dataset. We have explained the procedure of volume reduction in Figure 1. We have considered the average height of head around 300 mm for landmark detection purpose and cropped the volume for automatic landmark detection. The proposed method is based on the coarse to fine detection of cephalometric landmark. Therefore, two level of detection method was used.

During the coarse-to-fine detection, a landmark positional map was created for each cephalometric landmark. This map was created based on the average positional measurements of the landmarks. Before creating the landmark map, standard DL model VGG16 (Jiang et al., 2021; Tarhini

TABLE 1 20 cephalometric landmarks along with their definitions used in the study (Lindner et al., 2016).

ID	Landmarks' name	Abbreviation	Landmarks' definition
1	Sella	S	The geometric centre of the pituitary fossa as determined by inspection of a constructed point in the mid-sagittal plane.
2	Nasion	N	The intersection of the inter-nasal and fronto-nasal sutures, in the mid-sagittal plane.
3	Orbitale (Right)	OrR	The lowest-point on the lower edge of the right cranial orbit.
4	Orbitale (Left)	OrL	The lowest-point on the lower edge of the left cranial orbit
5	Frontozygomatic (right)	FzR	The most-medial and anterior point of right Frontozygomatic suture at the level of the lateral orbital rim
6	Frontozygomatic (left)	FzL	The most-medial and anterior point of left Frontozygomatic suture at the level of the lateral orbital rim
7	Zygion (right)	ZyR	The most-lateral point on the right outline of right zygomatic arch
8	Zygion (left)	ZyL	The most-lateral point on the left outline of left zygomatic arch
9	Condylion (right)	CoR	The most-superior point on the right mandibular condyle
10	Condylion (left)	CoR	The most-superior point on the left mandibular condyle
11	Basion	B	It is a median point of the anterior margin of the foramen magnum
12	A-point	A-pt	The deepest (most posterior) midline point on the curvature between the ANS and prosthion.
13	B-point	B-pt	The most-posterior midline point on the bony curvature of the anterior mandible, between infradentale and pogonion.
14	Pogonion	Pg	The most-anterior point on the contour of the bony chin.
15	Menton	Me	The most-inferior point of the mandibular symphysis.
16	Gnathion	Gn	The most-anterior inferior point on the bony chin.
17	Gonion (right)	GoR	The most-posterior inferior point on the outline of the angle of the right mandible.
18	Gonion (left)	GoL	The most-posterior inferior point on the outline of the angle of the left mandible.
19	Anterior nasal spine	ANS	The tip of the bony anterior nasal spine at the inferior margin of the piriform aperture.
20	Posterior nasal spine	PNS	The most posterior point on the bony hard palate (nasal floor).

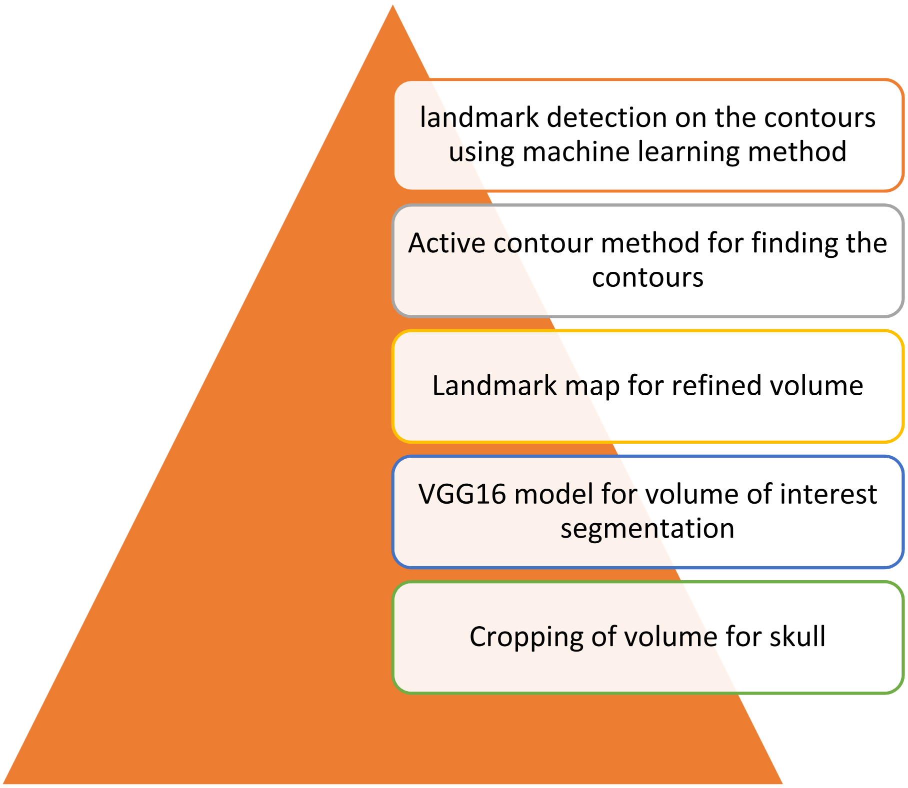

FIGURE 1 Systematically reducing the volume and obtaining volume of interest for 3D landmark detection.

et al., 2020) was used to automatic segment the volume of interest. This model was trained on 30 images only to determine that area of the volume is belonging to the skull or not. This method has reduced the undesirable volume significantly (Chamra & Harmanani, 2020; Hammoud et al., 2021; Sorkhoh et al., 2021). Then landmark map was used for the final volume of interest computation so that a particular landmark can be detected.

To create a landmark map, we have taken 30 sample Computational Tomography volumetric images and annotated the cephalometric landmarks manually as given in Table 1. The landmark positions of those 30 CT images were acquired and average volume was created for each landmark. This volume was considered as the volume where a landmark can exist on the new image. For the further variability of the landmark on the image we have taken an additional setoff of 5 mm at each boundary level to accommodate the landmark on new image. These volumes were considered as the landmark map volume for fine level detection. Active contour method was implemented to find the contours in each said volumes.

Geometric or geodesic active contour models (Caselles et al., 1997; Torosdagli et al., 2019) were used for the detection of boundary contours from the mapping volume. The active contour were implemented on the cropped volume slice by slice considering the 2D image rather than 3D image. The boundary contours as detected on each slice were combined to prepare the volumetric contours.

The volumetric contours were recorded because all the landmarks are defined on the contours only by referring their definition. A ML approach was used to find the relevant landmark over the existing volumetric contours. We have implemented a curve matching algorithm also to detect the precise landmark on the defined contours.

2.2 | Manual landmark annotations

To validate the proposed method, ground truths were necessary. We have requested three orthodontists to mark the cephalometric landmarks as shown in Table 1. For the marking of landmarks, Slicer software was used which allowed us to export the landmark coordinates in required format (see Figure 2). The average of each coordinate value was considered as the ground truth for the comparison of landmark coordinates. A total of 100 CT images were used for the validation purpose. However, the other 37 CT images were used for the training and volume map creation purposes. The orthodontists have annotated all the images with all the landmarks as given in Table 1.

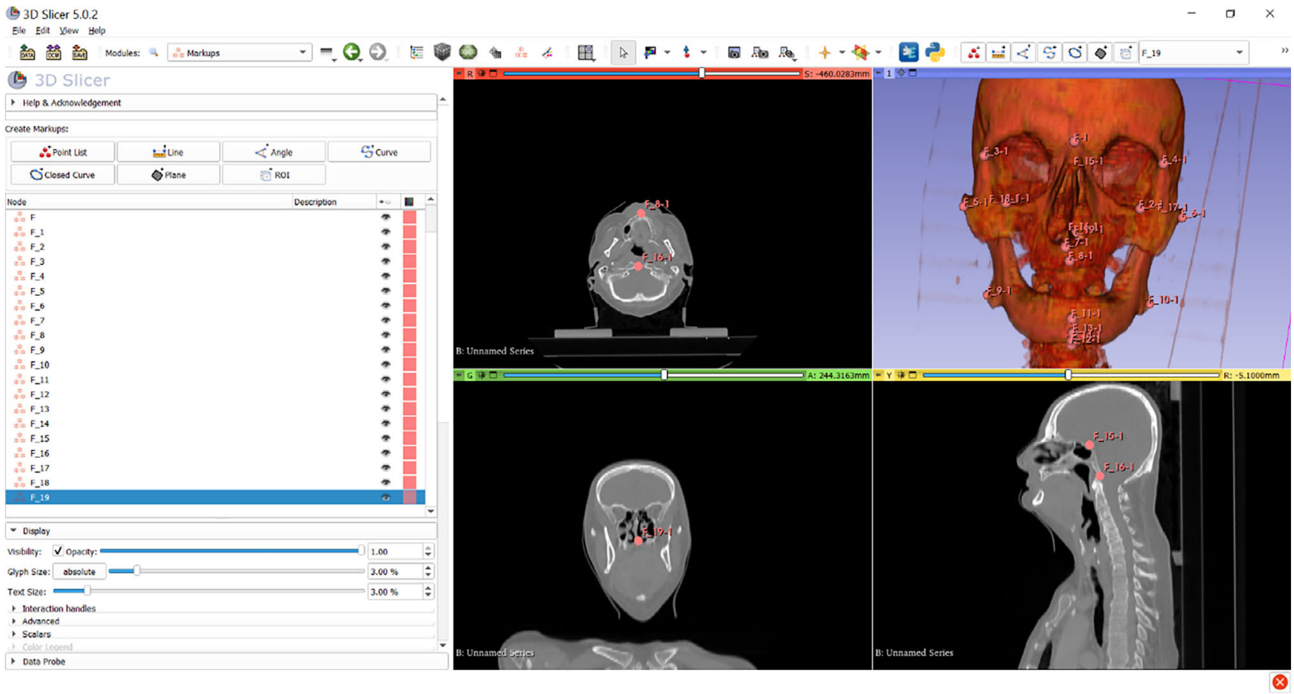

FIGURE 2 Manual landmark plotting on CT images using 3D Slicer software.

3 | RESULTS

To design the ground truth, three experts were requested to mark the 20 cephalometric landmarks. The Pearson's correlation was computed for all the measurements. The correlation coefficient was received as 0.91 for *X* axis, 0.92 for *Y* axis and 0.85 for *Z* axis. The correlation coefficient were found as good for considering those values as ground truths.

All the coordinate values as received from the proposed experiment were compared with the manual landmarks and the Euclidean distance was computed for each measurement on each image. For each landmark, the Euclidean distance was computed by considering the individual distances in all the three-axis and it was termed as detection error. The average error was computed for each cephalometric landmark by making a mean of the errors for the particular landmark. Similarly standard deviation was computed for each landmark. The mean (average) error, standard deviation and percentage detection rate for every landmark is listed in Table 2. The mean error for landmark detection was computed as 2.88 mm and standard deviation of 1.85 mm. According to Table 2, the results of detection rates can be considered. The highest detection rate was received as 100% for Nasion landmark for <4-mm error and the highest detection rate was received as 99% for Nasion landmark for <3-mm error.

4 | DISCUSSION

The artificial intelligence (AI) is more prevalent in the current era. The problems related to the orthodontic field are also being solved through AI and most specifically through the ML and DL (Abdellatef et al., 2022). There are several versions of AI for obtaining automation in the field of orthodontics and dentofacial deformities. The literature is available with different kinds of methods of AI named as convolutional neural network, neural network, supervised learning, unsupervised learning, You-Only-Look-Once, ML, DL and so on (Ashok & Gupta, 2021; Maken & Gupta, 2022; Pandey & Gupta, 2021). In the current article, we have proposed a hybrid ML and DL method is adopted to detect cephalometric landmark automatically.

The 3D automatic landmark detection was started in 2014 and still no promising method is available due to the complexity of the human anatomy. However, AI has made it little bit easy with the availability of the high computational power. But, AI requires higher amount of data for the training, and consequently large amount of training time is required. It raises the need of lightweight approaches to use the DL based approaches (Shen & Saab, 2021; Trivedi & Gupta, 2022). Therefore, initially, methods were not utilized using higher training data; although availability of the higher amount of data is also an issue. Table 3 shows the different similar methods which are available in the literature for the detection of 3D cephalometric landmarks. Starting with the model-based and knowledge-based, methods are converged with ML-based and DL-based

TABLE 2 Mean errors, standard deviation of the errors and the detection rate of cephalometric landmarks under 2, 3 and 4 mm range found in the experiment.

					Detection rate (%)		
ID	Name of the landmark	Abbreviation	Mean error	SD of the error	<2-mm	<3-mm	<4-mm
1	Sella	S	1.76	0.95	90	98	99
2	Nasion	N	1.27	0.77	91	99	100
3	Orbitale (Right)	OrR	3.89	2.86	68	73	85
4	Orbitale (Left)	OrL	4.02	3.14	64	70	80
5	Frontozygomatic (right)	FzR	3.78	2.34	70	72	75
6	Frontozygomatic (left)	FzL	3.96	2.19	67	79	80
7	Zygion (right)	ZyR	3.46	2.28	76	80	84
8	Zygion (left)	ZyL	4.12	3.48	71	75	80
9	Condylion (right)	CoR	3.91	2.78	48	55	72
10	Condylion (left)	CoR	3.12	2.04	45	55	70
11	Basion	B	1.84	1.08	80	83	90
12	A-point	A-pt	1.78	0.84	82	85	89
13	B-point	B-pt	2.14	0.98	80	87	88
14	Pogonion	Pg	2.45	1.85	80	85	90
15	Menton	Me	2.86	1.45	82	90	94
16	Gnathion	Gn	2.71	1.45	82	90	94
17	Gonion (right)	GoR	2.46	1.61	78	87	91
18	Gonion (left)	GoL	2.78	1.24	76	87	90
19	Anterior nasal spine	ANS	2.18	1.34	90	92	94
20	Posterior nasal spine	PNS	3.14	2.42	72	77	84

TABLE 3 Comparison of the landmark detection results with other methods found in the literature.

Sr. no.	Author year and reference	Number of landmarks	Testing sample size	Result
1.	Shahidi et al. (2014)	14	20	Detection rate as 63.57% under 3 mm error range
2.	Gupta et al. (2022)	20	30	Mean error = 2.01 mm; and detection rate as 82.67% under 3 mm error range
3.	Arisdakessian et al. (2020)	21	30	Maximum mean error in the measurements as 2.63 mm
4.	Montúfar et al. (2018)	18	24	Mean error 3.64 mm
5.	Neelapu et al. (2018)	20	30	Mean error 1.88 mm
6.	Lee et al. (2019)	7	7	Mean error 1.5 mm
7.	Yun et al. (2020)	93	26	Mean error 3.63 mm
8.	Kang et al. (2021)	16	8	Mean error 1.96 mm
9.	Dot et al. (2022)	33	38	95.4% detection rate under 3-mm error
10.	Proposed approach	20	100	Mean error 2.88 mm and 99% detection rate under 3 mm error range

methods. In that light, our method is also used hybrid ML and DL based method. By analysing Table 3, it was conceived that low number of testing sample size was taken for the experimentation initially which may not provide the required confidence for the obtained results. Therefore, the current experiment has taken 100 sample size for the testing of the proposed method. However, the required training at different steps was performed with lower number of data. The lower number of samples were taken for the training because we have also adopted the knowledge of the concern field. This available knowledge of the concern field helped to reduce the training data in the experimentation.

With the analysis of Table 3, it is clear that our mean error is higher than a few studies. However, our sample size is high which may give a confidence for the detection of the landmarks. The other studies like Lee et al. (2019) has shown least mean error as 1.5 mm but this error is demonstrated only on seven samples which may not be favourable for the other samples. In addition, if author applies the same method at large number of samples then the error may be increased. Therefore, our experiment is better than the literature methods in terms of higher amount of dataset used.

A drawback of our method is that we have used a dataset which includes a large amount of extra anatomical information which is not be relevant in the cephalometric analysis. This type of data was easily available on the web. Therefore, we have received the data and performed the experiments. Our proposed approach is new for its type as we have used ML and DL methods at different steps of the proposed methods. Our approach consists of the method with coarse-to-fine structure, which gives the opportunity to work in the structural manner to find the precise localization of the landmarks.

5 | CONCLUSION

We have proposed a hybrid DL and ML approach for the detection of cephalometric landmarks. It is the first of its kind of approach. The less number of datasets were used for the training and higher number of datasets were used for the testing. Compared to the literature methods, our method used higher number of datasets to demonstrate the accuracy of the proposed method. The mean error for landmark detection was computed as 2.88 mm and standard deviation of 1.85 mm. The highest detection rate was received as 100% for Nasion landmark for <4-mm error and the highest detection rate was received as 99% for Nasion landmark for <3-mm error.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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